

22/25

Name: Gaigi Suphalankar

Roll No: B027

SAPID: 60019220125

Branch: CSE(CIB)

## DE ASSIGNMENT 2

Q-1. Explain key engineering consideration for the ingestion phase.

⇒ ① Source variety:

- Different formats: Structured (tables), semi-structured (JSON, XML), unstructured (image, video, text)
- Different protocols: API calls, database connections, file transfers

② Data velocity

- Has fast data arriving (eg. daily batch vs real time system)
- Batch ingestion (periodic uploads) vs streaming ingestion (continuous) real time flow.

③ Data volume:

- How much data are you pulling? (MB, GB, TB - per day?)
- Systems need to handle large volumes without failure.

④ Fault tolerance and variability:

- What happens if ingestion fails? (eg network issue, partial loads)
- You must not lose data.

⑤ Data Quality & Validation

- Garbage in = garbage out.
- Validate data at ingestion point (eg. correct format)

Q-2. Discuss different ways to ingest data - direct database connection, change data capture and API's

⇒ ① Database connection: Direct database connections involve establishing a direct link between a data ingestion tool and a database. This method allows for immediate access to the data within the database, making it suitable for applications that require real time data access.

- JDBC (Java Database Connectivity): A standard API for Java application to connect to database.
- ODBC (Open database connectivity): A standard API for Java application to connect to database.
- Data-specific API's: Many databases offer their own API's (e.g. MySQL connector, postgresql) for direct connection.
- Data virtualization: Techniques like data virtualization allow connecting to multiple data sources through single interface.
- ② Change Data Capture: It is a software process that identifies and tracks changes to data in a database.
  - Audit columns: By using "LAST-UPDATED" or "DATE-MODIFIED" columns, by adding them if not in the application.
  - Step 1: Get the max value of both the target tables 'created time' and 'updated-time' columns.
  - Step 2: Select the rows with 'created time' greater than (2) the target table's max 'created time'.
- Table deltas: You can use table delta or 'table-diff' utilities to compare the data in two tables for non convergence. There are several examples of SQL scripts that can find the difference of two tables.
- Trigger based CDC: Defining database triggers and creating your own change log in shadow tables. Triggers fire before or after INSERT, UPDATE or DELETE commands operating at the level, some users prefer this approach. Some databases even have native support for triggers.

- Q3. Explain data transformations as a part of pipeline with example.
- ⇒ ① Data cleaning: Fixes issues like missing values, duplicates and inconsistencies.

Example: Replacing ~~Attt~~ <sup>null</sup> user email addresses with a placeholder or removing duplicate transactions.

- ② Data Validation: Ensures data meets predefined rules and constraints.

Example: Checking that all customers follow a specific format and are unique.

- ③ Data Formatting: Converts data type and target system requirements.

Example: Standardization of date formats (DD/MM/YYYY) → (YYYY-MM-DD) and converting string to integers.

- ④ Data Enrichment: Add new, useful information from external sources or calculations.

Example: Appending demographic data based on ZIP codes or calculating customer lifetime value.

- ⑤ Data Aggregation: Combines data into summary metrics for analysis.

Example: Summing all purchases made by a customer in a month.

- ⑥ Data Normalization: Standard data for consistency across the dataset.

Example: Converting all currency values to users or standardize test cases.

Q.4. Discuss data preparation transformation with optimizing the file format, optimizing with data partitioning, data cleaning.

→ Optimizing the file format:

- Choose an efficient file format improves storage, usage, read/write speed and overall performance. Traditional formats like csv and json are simple but inefficient for large datasets analytics.

- Columnar formats like Parquet and ORC store data by column enabling faster query performance when accessing only selected columns.
- ~~Also~~ Arrow is suitable for row-based processing and streaming data and it supports schema evolution.
- Optimizing with data partitioning.
- Partitioning breaks datasets into smaller segments. Helps engines like Spark, Hive read only relevant data, improving query importance.
- Example: A sales dataset partitioned by year and month allow January 2024 queries to access just that partition.
- Benefits of data partitioning:
  - Improve scalability
  - Improve Performance
  - Improve Security
  - Improve Availability.
- Data Cleansing
- Cleansing ensures that the data is accurate, consistent and usable. Common issues include missing values, duplicates, inconsistent formats, typo, etc.
- Cleansing Techniques:
  - 1) Handling missing data.
  - 2) Remove duplicates
  - 3) Type and range validation

Q.5. Explain in detail: Active security and the principle of least privilege.

- (i) Active security: Refers to the real time protection mechanisms that monitor, detect and respond to cybersecurity threats activity, rather than relying solely on predefined

rules or passive defenses.

Key features:

① → Real time monitoring: Continuously watches system activity, networks traffic and user behaviour.

② → Automated threat response: Quickly blocks, disables suspected malicious activity through automated system.

③ Principle of Least Privilege (PoLP): A fundamental cybersecurity concept where users and systems are granted the minimum level of access rights required to perform main functions.

Implementation:

→ Assign roles with only necessary permissions.

→ Regular audit user roles and access rights

→ Use time bound access controls for temporary tasks.

Q.6. Explain in detail cataloguing your data to avoid data Swamp.

⇒ ① Data Swamp:

A data swamp is a poorly managed data value that lacks structure, metadata and governance, making it difficult to find trust and one data.

• Unlike a well organized data lake, a data swamp leads to data chaos where datasets are inaccessible, redundant, outdated and low quality.

② Importance of data Cataloguing:

• It is a process of creating an organized inventory of data assets including metadata, lineage and relationship between datasets.

• It acts like a library index, helping users discover, understand and use the right data efficiently.

### ⑤ Features of data cataloguing

- Metadata management
- Search and Discovery.
- Data lineage tracing.

### ⑥ How cataloguing prevents a data swamp.

- Improves visibility; Users can find and reuse datasets rather than duplicating efforts.
- Enables governance: Ensures policies, access controls and data standards are enforced.
- Enhance data quality: Promote documentation, validation and consistent use of datasets.
- Support Compliance: Helps organizations meet data protection and audit requirements.